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Experiment 2: Containerizing Applications Using Docker

Aim:

To understand the fundamentals of containerization and to perform basic container operations using Docker.

This experiment maps to LO3.

LO3: Apply containerization tools such as Docker to package, distribute and run applications in isolated, reproducible environments.

Theory:

Containerization is an operating-system-level virtualization technique in which an application and everything it depends on — libraries, binaries, and configuration files — are packaged together into a single portable unit called a container. Unlike a virtual machine, a container does not carry a full guest operating system; it shares the kernel of the host machine and is kept separate from other containers by kernel features such as namespaces and control groups (cgroups). This makes containers far lighter, faster to start, and more resource-efficient than virtual machines.

Docker is the most widely used containerization platform. It follows a client–server architecture in which the Docker CLI issues commands to the Docker daemon, which then builds, runs, and manages containers. Its two central concepts are the image and the container: an image is a read-only, layered template describing an application's filesystem and metadata, while a container is a running, writable instance of that image. Images are distributed through registries such as Docker Hub, from which they can be searched and pulled. Because an image captures the exact runtime environment, the same image behaves identically on any host running a Docker engine, which resolves the familiar “it works on my machine” problem and makes Docker a foundational tool in modern DevOps and CI/CD pipelines.

Implementation:

1. Verify the Docker Installation:

Prints the installed Docker Engine version and build number. This confirms that the Docker client is installed correctly and reachable on the system PATH before any container operation is attempted.

```
Last login: Sun Jul 19 22:08:32 on ttys000
dhruv@Dhruvs-MacBook-Air ~ % docker --version
Docker version 29.4.3, build 055a478
```

2. List Local Images:

Lists every image currently held in the local image cache together with its ID, disk usage, and content size. At this stage only three images are present, all pulled during earlier work.

```
dhruv@Dhruvs-MacBook-Air ~ % docker images
                                                  i Info →  U  In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
pihole/pihole:latest                 70a9c11171de       164MB         43.3MB    U
postgres:16                          17e67d7b9890       663MB          165MB
timescale/timescaledb:2.17.2-pg16
                                     4e459e217f00      2.74GB          716MB    U
```

3. Check the Search Syntax:

Running docker search without a search term returns a usage error. Docker expects exactly one TERM argument, and the error output documents the correct syntax for the command.

```
dhruv@Dhruvs-MacBook-Air ~ % docker search
docker: 'docker search' requires 1 argument

Usage:  docker search [OPTIONS] TERM

Run 'docker search --help' for more information
```

4. Search Docker Hub for MySQL Images:

Queries the Docker Hub registry for repositories whose name or description matches mysql. Results are ranked by star count, and the OFFICIAL column marks the image maintained by the upstream project rather than by the community.

```
dhruv@Dhruvs-MacBook-Air ~ % docker search mysql
NAME                     DESCRIPTION                                     STARS     OFFICIAL
mysql                    MySQL is a widely used, open-source relation…   16157     [OK]
circleci/mysql           MySQL is a widely used, open-source relation…   33
bitnamicharts/mysql      Bitnami Helm chart for MySQL                    1
cimg/mysql                                                               4
ubuntu/mysql             MySQL open source fast, stable, multi-thread…   74
google/mysql             MySQL server for Google Compute Engine          26
elestio/mysql            Mysql, verified and packaged by Elestio         4
linuxserver/mysql        A Mysql container, brought to you by LinuxSe…   49
alpine/mysql             mysql client                                    8
docksal/mysql            MySQL service images for Docksal - https://d…   0
eclipse/mysql            Mysql 5.7, curl, rsync                          3
cleanstart/mysql         Secure by Design, Built for Speed, Hardened …   0
ilios/mysql              Mysql configured for running Ilios              1
datajoint/mysql          MySQL image pre-configured to work smoothly …   2
ddev/mysql               ARM64 base images for ddev-dbserver-mysql-8.…   2
mysql/mysql-server       Optimized MySQL Server Docker images. Create…   1035
mirantis/mysql                                                           0
cbioportal/mysql         This repository hosts MySQL database images …   1
corpusops/mysql          https://github.com/corpusops/docker-images/     0
vulhub/mysql                                                             2
vitess/mysql             Lightweight image to run MySQL with Vitess      1
mysql/mysql-router       MySQL Router provides transparent routing be…   29
mysql/mysql-cluster      Experimental MySQL Cluster Docker images. Cr…   101
nasqueron/mysql                                                          1
mysql/mysql-operator     MySQL Operator for Kubernetes                   1
```

5. Search Using a Misspelled Term:

Searching for a misspelled term returns only unrelated community images with almost no stars. This shows that docker search performs plain substring matching on repository names and does not correct spelling.

```
dhruv@Dhruvs-MacBook-Air ~ % docker search ubantu
NAME                      DESCRIPTION          STARS     OFFICIAL
boystar/ubantu            Ubantu系统精简版      4
```

6. Search for Nginx Images:

Returns the official Nginx image alongside numerous vendor and community variants such as nginx/nginx-ingress and linuxserver/nginx, illustrating how many specialised builds a single project can have on the public registry.

```
dhruv@Dhruvs-MacBook-Air ~ % docker search nginx
docker search alpine
docker search postgres
NAME                                      DESCRIPTION                                     STARS     OFFICIAL
nginx                                     Official build of Nginx.                        21340     [OK]
nginx/nginx-ingress                       NGINX and  NGINX Plus Ingress Controllers fo…  122
nginx/nginx-prometheus-exporter           NGINX Prometheus Exporter for NGINX and NGIN…  52
nginx/nginx-ingress-operator              NGINX Ingress Operator for NGINX and NGINX P…  4
nginx/nginxaas-loadbalancer-kubernetes                                                    1
bitnamicharts/nginx                       Bitnami Helm chart for NGINX Open Source        4
ubuntu/nginx                              Nginx, a high-performance reverse proxy & we…  141
kasmweb/nginx                             An Nginx image based off nginx:alpine and in…  9
rancher/nginx                                                                             4
linuxserver/nginx                         An Nginx container, brought to you by LinuxS…  236
dtagdevsec/nginx                          T-Pot Nginx                                     0
vmware/nginx                                                                              3
paketobuildpacks/nginx                                                                    0
gluufederation/nginx                       A customized NGINX image containing a consu…  1
cleanstart/nginx                          Secure by Design, Built for Speed, Hardened …  0
activestate/nginx                         ActiveState's customizable, low-to-no vulner…  0
antrea/nginx                              Nginx server used for Antrea e2e testing        0
intel/nginx                                                                               0
docksal/nginx                             Nginx service image for Docksal                 1
geokrety/nginx                            Our customized nginx image                      0
circleci/nginx                            This image is for internal use                  2
ilios/nginx                               Nginx customized to run Ilios along with the…  0
corpusops/nginx                           https://github.com/corpusops/docker-images/     1
dockette/nginx                            Nginx SSL / HSTS / HTTP2                        3
wayofdev/nginx                            Nginx Docker image for PHP development. SSL-…  0
```

7. Search for Alpine Images:

Alpine Linux is a minimal distribution of roughly five megabytes that is widely used as a lightweight base image. The results list the official image together with many small purpose-built tool images derived from it.

```
NAME                DESCRIPTION                                     STARS     OFFICIAL
alpine              A minimal Docker image based on Alpine Linux…   11536     [OK]
alpine/git          A  simple git container running in alpine li…  252
alpine/socat        Run socat command in alpine container           118
alpine/curl                                                         13
alpine/helm         Auto-trigger docker build for kubernetes hel…   70
alpine/k8s          Kubernetes toolbox for EKS (kubectl, helm, i…   67
alpine/bombardier   Auto-trigger docker build for bombardier whe…   28
alpine/httpie       Auto-trigger docker build for `httpie` when …   22
alpine/terragrunt   Auto-trigger docker build for terragrunt whe…   19
alpine/openssl      openssl                                         8
alpine/kubectl      Kubernetes command-line tool for managing cl…   1
alpine/psql         psql — The PostgreSQL Command-Line Client       4
alpine/flake8       Auto-trigger docker build for fake8 via ci c…   2
alpine/openclaw     OpenClaw - Your own personal AI assistant. A…   134
alpine/ansible      run ansible and ansible-playbook in docker      41
alpine/jmeter       run jmeter in Docker                            9
alpine/java         Repo containing the build scripts to produce…   5
alpine/semver       Docker tool for semantic versioning             4
alpine/xml          several xml tools to work on xml file as jq …   2
alpine/mongosh      mongosh - MongoDB Command Line Database Tools   3
alpine/mysql        mysql client                                    8
alpine/ollama       Minimal CPU-only Ollama Docker Image            20
alpine/crane                                                        0
alpine/cfn-nag      Auto-trigger docker build for cfn-nag when n…   0
alpine/gcloud       Auto-trigger docker build for gcloud (google…   0
```

8. Search for PostgreSQL Images:

Lists the database images published under the postgres keyword. Comparing star counts makes it straightforward to identify the trustworthy official image among many similarly named alternatives.

```
NAME                      DESCRIPTION                                     STARS     OFFICIAL
postgres                  The PostgreSQL object-relational database sy…   14969     [OK]
cimg/postgres                                                             9
circleci/postgres         The PostgreSQL object-relational database sy…   35
kasmweb/postgres          Postgres image maintained by Kasm Technologi…   6
elestio/postgres          Postgres, verified and packaged by Elestio      2
ubuntu/postgres           PostgreSQL is an open source object-relation…   45
cleanstart/postgres       Secure by Design, Built for Speed, Hardened …   0
artifacthub/postgres                                                      0
geokrety/postgres         Postgres with postgis + quantile and amqp ex…   0
corpusops/postgres        https://github.com/corpusops/docker-images/     0
rootpublic/postgres                                                       0
dockette/postgres         My PostgreSQL image with tunning and preinst…   1
vulhub/postgres                                                           2
wayofdev/postgres                                                         0
pgrouting/postgres         Postgres Docker images with PostGIS and dep…   0
uselagoon/postgres                                                        0
openeuler/postgres                                                        0
clarinpl/postgres                                                         0
mosipdev/postgres                                                         0
supabase/postgres         Unmodified Postgres with some useful plugins…   78
brimstone/postgres        postgres image with traefik-cert support        0
trainlineeurope/postgres  Extended version of official Postgres https:…   0
blacklabelops/postgres    Postgres Image for Atlassian Applications       4
fredboat/postgres         PostgreSQL 10.0 used in FredBoat's docker-co…   1
sushantdhiman/postgres    Postgres Image for Sequelize                    2
```

9. Search Using a Tagged Name:

docker search matches repository names only, so appending a tag does not filter the result set. The command still returns the full list of ubuntu repositories, confirming that tags are resolved when an image is pulled rather than when it is searched.

```
dhruv@Dhruvs-MacBook-Air ~ % docker search ubuntu:latest
NAME                     DESCRIPTION                                     STARS     OFFICIAL
ubuntu                   Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid             Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx             Nginx, a high-performance reverse proxy & we…   141
ubuntu/kafka             Apache Kafka, a distributed event streaming …   62
ubuntu/apache2           Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus        Prometheus is a systems and service monitori…   80
ubuntu/jre               Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/bind9             BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/postgres          PostgreSQL is an open source object-relation…   45
ubuntu/zookeeper         ZooKeeper maintains configuration informatio…   15
ubuntu/dotnet-aspnet     Chiseled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python            A chiseled Ubuntu rock with the Python runt…   33
ubuntu/mysql             MySQL open source fast, stable, multi-thread…   74
ubuntu/dotnet-deps       Chiseled Ubuntu for self-contained .NET & A…   16
ubuntu/dotnet-runtime    Chiseled Ubuntu runtime image for .NET apps…   24
ubuntu/redis             Redis, an open source key-value store. Long-…   26
ubuntu/grafana           Grafana, a feature rich metrics dashboard & …   12
ubuntu/jdk               Ubuntu rock for OpenJDK                         0
ubuntu/memcached         Memcached, in-memory keyvalue store for smal…   7
ubuntu/rabbitmq          Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/alertmanager      Ubuntu Rock for Alertmanager. It handles ale…   1
ubuntu/mimir             Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/grafana-agent     Ubuntu ROCK for Grafana Agent, an open-sourc…   0
ubuntu/node              Ubuntu-based Node.js image for server-side a…   5
ubuntu/loki              Grafana Loki, a log aggregation system like …   2
```

10. Verify the Downloaded Images:

After pulling images from the registry, the local cache now holds seven images. The U marker in the EXTRA column flags those that are currently in use by a running container.

```
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest                        28bd5fe8b56d       13.6MB         4.27MB
hello-world:latest                   c3cbe1cc1aa5       22.6kB         10.3kB
nginx:latest                         5a88c9c45479        259MB         64.3MB
pihole/pihole:latest                 70a9c11171de        164MB         43.3MB    U
postgres:16                          17e67d7b9890        663MB          165MB
timescale/timescaledb:2.17.2-pg16    4e459e217f00       2.74GB          716MB    U
ubuntu:latest                        3131b4cc82a7        180MB         44.4MB
```

11. Run an Interactive Container:

Creates and starts a new container from the ubuntu image. The -i flag keeps standard input open and -t allocates a pseudo-terminal, which together provide an interactive shell inside the container. The changed prompt, root@f016f67c9267, shows the container ID and confirms that the session is now running inside the container.

```
dhruv@Dhruvs-MacBook-Air ~ % docker run -it ubuntu
root@f016f67c9267:/# cat /etc/os-release
uname -a
pwd
ls
whoami
```

12. Identify the Container Operating System:

Reads the OS identification file inside the container, which reports Ubuntu 26.04 LTS. The host machine is macOS, so this demonstrates that a container supplies a complete Linux userland that is entirely independent of the host operating system.

```
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
```

13. Inspect the Kernel:

Prints the running kernel version. The kernel reported is 6.12.76-linuxkit, which belongs to the underlying Docker virtual machine rather than to the container itself — direct evidence that containers share a kernel instead of booting one of their own.

```
Linux f016f67c9267 6.12.76-linuxkit #1 SMP Fri May  1 14:35:41 UTC 2026 aarch64 GNU/Linux
```

14. Examine the Container Filesystem:

The shell starts in the root directory, and listing it reveals a complete Linux filesystem hierarchy — bin, etc, home, usr, var and the rest — supplied entirely by the layers of the image.

```
/
bin  boot  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
```

15. Check the Current User:

Confirms that the default user inside the container is root, which is why packages can be installed later without invoking sudo.

```
root
root@f016f67c9267:/#
```

16. Navigate Inside the Container:

Changing directory and printing the working directory behave exactly as they would on an ordinary Linux host, confirming that the container presents a fully navigable and writable filesystem.

```
root@f016f67c9267:/# cd /etc
root@f016f67c9267:/etc# pwd
/etc
```

17. Inspect the Container Process Table:

Lists the running processes. Only two appear: /bin/bash running as PID 1 and the ps command itself. Process-ID namespacing hides every process belonging to the host, so the container observes an isolated process tree in which its own shell acts as the init process.

```
root@f016f67c9267:/etc# ps -ef
UID        PID  PPID  C STIME TTY          TIME CMD
root         1     0  0 05:31 pts/0    00:00:00 /bin/bash
root        14     1  0 05:49 pts/0    00:00:00 ps -ef
```

18. Observe a Missing Utility:

The pstree command fails with “command not found”. Official base images ship with only a minimal package set in order to keep image size small, so any additional tooling has to be installed explicitly.

```
root@f016f67c9267:/etc# pstree
bash: pstree: command not found
```

19. Update Package Lists and Install a Package:

Refreshes the package index from the Ubuntu ports repositories and installs psmisc, the package that provides pstree. The arm64 architecture in the output matches the Apple Silicon host, showing that the image pulled was built for the host's CPU architecture.

```
bash: pstree: command not found
root@f016f67c9267:/etc# apt update && apt install -y psmisc
Get:1 http://ports.ubuntu.com/ubuntu-ports resolute InRelease [136 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports resolute-updates InRelease [137 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports resolute-backports InRelease [136 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports resolute-security InRelease [137 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports resolute/main arm64 Packages [1860 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports resolute/multiverse arm64 Packages [298 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports resolute/universe arm64 Packages [19.9 MB]
Get:8 http://ports.ubuntu.com/ubuntu-ports resolute/restricted arm64 Packages [231 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports resolute-updates/main arm64 Packages [466 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports resolute-updates/multiverse arm64 Packages [3592 B]
Get:11 http://ports.ubuntu.com/ubuntu-ports resolute-updates/restricted arm64 Packages [426 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports resolute-updates/universe arm64 Packages [256 kB]
Get:13 http://ports.ubuntu.com/ubuntu-ports resolute-security/restricted arm64 Packages [413 kB]
Get:14 http://ports.ubuntu.com/ubuntu-ports resolute-security/main arm64 Packages [391 kB]
Get:15 http://ports.ubuntu.com/ubuntu-ports resolute-security/universe arm64 Packages [168 kB]
Fetched 25.0 MB in 5s (5178 kB/s)
1 package can be upgraded. Run 'apt list --upgradable' to see it.
Installing:
```

20. Complete the Installation:

The package is unpacked and configured inside the container's writable layer. These changes exist only within this particular container and are lost once it is removed, unless they are committed to a new image or declared in a Dockerfile.

```
Installing:
  psmisc

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
  Download size: 179 kB
  Space needed: 864 kB / 223 GB available

Get:1 http://ports.ubuntu.com/ubuntu-ports resolute/main arm64 psmisc arm64 23.7-2ubuntu2 [179 kB]
Fetched 179 kB in 1s (143 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79, <STDIN> line 1.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/aarch64-linux-gnu/perl/5.40.1 /usr
/local/share/perl/5.40.1 /usr/lib/aarch64-linux-gnu/perl5/5.40 /usr/share/perl5 /usr/lib/aarch64-linux-gnu/perl-base /usr/lib/aarch64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/lo
cal/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8, <STDIN> line 1.)
debconf: falling back to frontend: Teletype
Selecting previously unselected package psmisc.
(Reading database ... 7720 files and directories currently installed.)
Preparing to unpack .../psmisc_23.7-2ubuntu2_arm64.deb ...
```

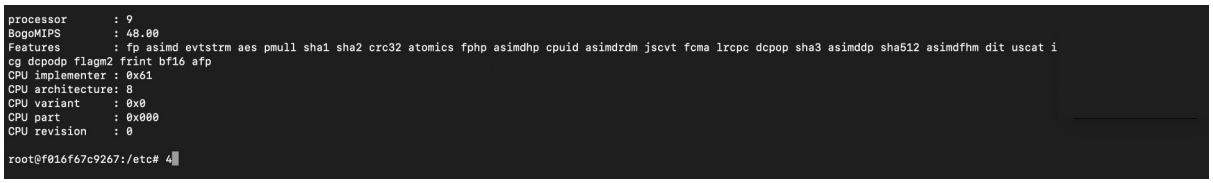
21. Inspect the Allocated CPU Resources:

Displays the processors visible to the container. Each entry describes an ARM (aarch64) core along with the instruction-set features it supports, exposed through the Docker virtual machine.

```
Setting up psmisc (23.7-2ubuntu2) ...
root@f016f67c9267:/etc# cat /proc/cpuinfo
processor       : 0
BogoMIPS        : 48.00
Features        : fp asimd evtstrm aes pmull sha1 sha2 crc32 atomics fphp asimdhp cpuid asimdrdm jscvt fcma lrcpc dcpop sha3 asimddp sha512 asimdfhm dit uscat ilrcpc flagm sb paca pa
cg dcpodp flagm2 frint bf16 afp
CPU implementer : 0x61
CPU architecture: 8
CPU variant     : 0x0
CPU part        : 0x000
CPU revision    : 0
```

22. Confirm the Full CPU Allocation:

The listing ends at processor 9, showing that ten cores are visible to the container. Docker does not virtualize CPUs; the container simply sees the host's cores and is restricted only by cgroup limits, which are unrestricted by default.



Conclusion:

In this experiment, the fundamental workflow of running containerized applications with Docker was carried out successfully. The environment was first verified with docker --version, and the local image cache was inspected using docker images. The Docker Hub registry was then explored with docker search across several terms — mysql, nginx, alpine, postgres, and ubuntu — which demonstrated how official images are distinguished from community images, how star counts serve as an indicator of reliability, and how the search command matches repository names by plain substring rather than by tag.

An interactive Ubuntu container was then launched with docker run -it, and its runtime environment was examined from the inside. Inspecting /etc/os-release, the kernel version, the root filesystem, the process table, and /proc/cpuinfo confirmed the defining characteristics of containerization: the container provides a complete and isolated Ubuntu userland with its own filesystem and process namespace, yet shares the host's linuxkit kernel and CPU cores instead of emulating hardware the way a virtual machine would. Installing psmisc inside the running container further showed that the writable container layer can be modified at runtime, while highlighting that such modifications are ephemeral and are better declared in an image definition for reproducibility. Overall, the experiment provided a practical understanding of images, containers, registries, and kernel-level isolation — the concepts that underpin container-based application deployment in modern DevOps practice.